

Method	ANM-HSIC	IGCI	RCC	CGAN-C2ST	Ensemble	C2ST type
Accuracy	67%	71%	76%	73%	82%	KNN
				70%	73%	NN
				58%	65%	MMD

Table 4: Results on cause-effect discovery on the Tübingen pairs experiment.

The need to ensemble is a remainder of the unstable behaviour of generative adversarial training, but also highlights the promise of such models for causal discovery.

7 CONCLUSION

Our *take-home message* is that modern binary classifiers can be easily turned into powerful two-sample tests. We have shown that these *classifier two-sample tests* set a new state-of-the-art in performance, and enjoy unique attractive properties: they are easy to implement, learn a representation of the data on the fly, have simple asymptotic distributions, and allow different ways to interpret how the two samples under study differ. Looking into the future, the use of binary classifiers as two-sample tests provides a flexible and scalable approach for the evaluation and comparison of generative models (such as GANs), and opens the door to novel applications of these methods, such as causal discovery.

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